

KHOI NGUYEN

Electrical Engineering Student — Digital ASIC, VLSI & Physical Design

+1 431-374-5778 | anhkhoi_2005@icloud.com | [linkedin.com/in/anhkhoilenguyen](https://www.linkedin.com/in/anhkhoilenguyen) | aakn0401.github.io/portfolio

Experience

University of Waterloo

April 2026–Sept 2026

ASIC Design Co-op / Visiting Research Fellow – CMOS Design and Reliability Group

Waterloo, ON

- Owned backend physical design for a multi-block RISC-V SoC in GlobalFoundries 12LP+, driving synthesis, floorplanning, and timing closure in Cadence Genus/Innovus across concurrent design blocks.
- Completed full RTL-to-GDSII implementation and tapeout across the Cadence digital flow (Genus, Innovus, Joules, Pegasus, Modus), achieving signoff closure for chip submission to fabrication.
- Built and validated production-grade DFT infrastructure in Cadence Modus – scan insertion, ATPG pattern generation, and fault-simulation debug – meeting test coverage sign-off criteria used for tapeout readiness.
- Designed and verified standard-cell layouts (DFFs, SRAM bitcells) in 12nm FinFET using Cadence Virtuoso, cutting cell area by 10% while holding DRC/LVS clean.

Nano-Systems Fabrication Laboratory (Class 10 Cleanroom)

May 2025–April 2026

Student Researcher

University of Manitoba

- Executed end-to-end CMOS and MEMS fabrication in a Class 10 cleanroom, including 200 nm DUV photolithography, ion implantation, RIE/ICP etching, and PVD/CVD thin-film deposition.
- Developed and optimized 200 nm DUV lithography processes (alignment, exposure dose, focus tuning), achieving high pattern fidelity and sub-micron critical dimension (CD) control.
- Optimized plasma etch and lithography recipes, reducing defect rate by 30–50% and improving etch throughput by 40%.

Projects

NVIDIA NVDLA AI Accelerator: Full-Chip Implementation in 12-nm FinFET | *Cadence Genus, Modus, Innovus*

- Implemented the open-source NVIDIA NVDLA accelerator RTL-to-GDSII in 12-nm FinFET using a hierarchical 6-partition flow (6.9M instances), closing timing at 1.11 GHz across 2 clock domains at the slow corner with zero negative slack over 3.5M reg-to-reg paths.
- Inserted integrated clock-gating cells (ICGs) during Genus synthesis to cut dynamic clock-tree power for a total power of 7.8 W; balanced gated clocks during CTS in Innovus.
- Generated 128 dual-port SRAM hard macros (RVT/MVT) with a foundry memory compiler, replacing behavioral RAM models to enable synthesis and P&R
- Completed placement, CTS, and routing in Innovus at 62.75% cell density, reaching signoff-clean physical verification with 0 DRC violations and 0 connectivity errors.

Adaptive DVFS RISC-V CPU with Timing-Error Detection in 12nm FinFET | *Verilog, Cadence Genus/Innovus/Modus*

- Architected timing-error detection flip-flops to enable adaptive dynamic voltage-frequency scaling (DVFS), targeting energy efficiency gains under variable workload.
- Engineered custom scan chain/DFF architectures across 6 fault-detection schemes and built full DFT infrastructure in Cadence Modus, achieving 90%+ combined static and dynamic fault coverage.
- Closed timing at 1 GHz with zero violations across 4,739 paths via RTL synthesis in Cadence Genus using MMMC constraints and full LEF/technology library integration.
- Executed full-chip implementation in Cadence Innovus, including floorplanning, power planning, placement, CTS.
- Delivered a 9,070.45 μm^2 implementation with 15,319 leaf instances, from RTL to GDSII, in GlobalFoundries 12LP+.

Electronics-Printed MEMS Lorentz Relay with Electrostatic Hold-Down | *COMSOL, Voltera NOVA, RIE*

- Designed a hybrid MEMS relay combining Lorentz-force actuation for fast switching with electrostatic hold-down to latch the closed state without continuous current.
- Fabricated the device on flexible polyimide using additive printed electronics (Voltera NOVA) and released suspended structures with O_2 reactive ion etching.
- Characterized the relay on a probe station, measuring 109.8 $\mu\text{m}/\text{mA}$ actuation sensitivity and a 25 V minimum electrostatic hold-down voltage; presented at the Undergraduate Research Showcase (3rd place).

Technical Skills

Digital ASIC and Physical Design: Verilog, RISC-V architecture, Cadence Genus, Innovus, synthesis, floorplanning, place and route (PnR), clock tree synthesis (CTS), static timing analysis (STA), Design for Test (DFT), ATPG

Analog & Mixed-Signal IC Design: Cadence Virtuoso, ADE/Spectre, nanoscale CMOS design (12 & 14 nm FinFET, 12 nm FDSOI, 45 nm, 65 nm), full-custom layout techniques, parasitic extraction, DRC/LVS verification

Programming and Systems: Python, TCL, MATLAB, Linux/Unix, ARM assembly

Education

University of Manitoba

Aug 2023–Dec 2027

Bachelor of Science in Electrical Engineering (GPA: 4.19/4.5)

Winnipeg, Canada

Awards: Undergraduate Research Award (2025), Dean's Honour List (2023–2026), 3rd Place Undergraduate Research (2025)